1. Deploy your your previous assignment flask frontend and express backend in kubernetes cluster locally with minikube.

HERE IS THE OUTPUTS:

Folder Structure of my project (K8s-assignment)

```
K8s-assignment/
│
├── flask-frontend/
│   ├── app.py
│   ├── requirements.txt
│   ├── templates/
│   │   └── index.html
│   └── Dockerfile
│
├── python-backend/
│   ├── server.py
│   ├── requirements.txt
│   └── Dockerfile
│
├── k8s/
│   ├── flask-deployment.yaml
│   ├── backend-deployment.yaml
│   ├── mongo-deployment.yaml
│   ├── mongo-express-deployment.yaml
│
└── README.md
```

INSTALLATION OF MINIKUBE IN LINUX:

Install dependencies
CMD : sudo apt-get update -y

CMD : sudo apt-get install -y curl apt-transport-https virtualbox virtualbox-ext-pack

Install kubectl
CMD : curl -LO https://storage.googleapis.com/kubernetes-release/release/`curl -s https://storage.googleapis.com/kubernetes-release/release/stable.txt`/bin/linux/amd64/kubectl

CMD : chmod +x kubectl

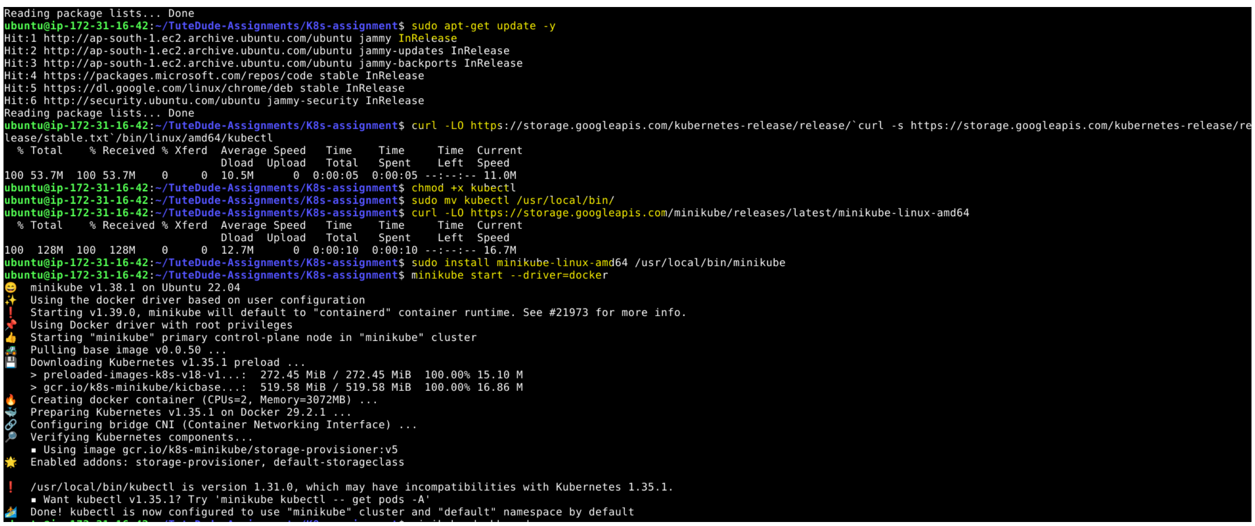
CMD : sudo mv kubectl /usr/local/bin/

Install minikube
CMD : curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64

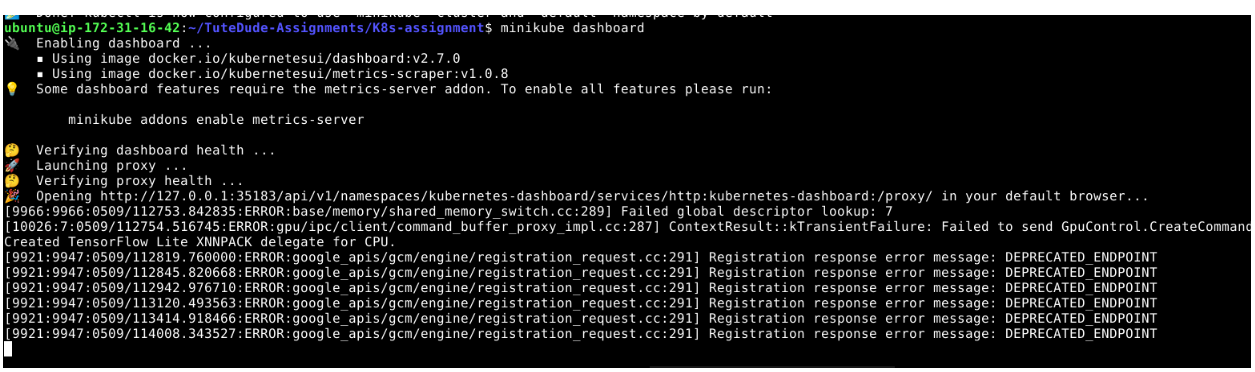
CMD : sudo install minikube-linux-amd64 /usr/local/bin/minikube

Start cluster
CMD : minikube start --driver=docker

Enable dashboard
CMD : minikube dashboard



```
Reading package lists... Done
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ sudo apt-get update -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.microsoft.com/repos/code stable InRelease
Hit:5 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:6 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ curl -LO https://storage.googleapis.com/kubernetes-release/release/`curl -s https://storage.googleapis.com/kubernetes-release/re
lease/stable.txt`/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 53.7M  100 53.7M    0     0  10.5M      0  0:00:05  0:00:05 --:--:-- 11.0M
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ chmod +x kubectl
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ sudo mv kubectl /usr/local/bin/
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  128M  100  128M    0     0  12.7M      0  0:00:10  0:00:10 --:--:-- 16.7M
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ minikube start --driver=docker
😄  minikube v1.38.1 on Ubuntu 22.04
✨  Using the docker driver based on user configuration
❗  Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
📌  Using Docker driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 15.10 M
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 16.86 M
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.35.1.
    ▪ Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```



```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment$ minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:35183/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
[9966:9966:0509/112753.842835:ERROR:base/memory/shared_memory_switch.cc:289] Failed global descriptor lookup: 7
[10026:7:0509/112754.516745:ERROR:gpu/ipc/client/command_buffer_proxy_impl.cc:287] ContextResult::kTransientFailure: Failed to send GpuControl.CreateCommand
Created TensorFlow Lite XNNPACK delegate for CPU.
[9921:9947:0509/112819.760000:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9921:9947:0509/112845.820668:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9921:9947:0509/112942.976710:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9921:9947:0509/113120.493563:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9921:9947:0509/113414.918466:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9921:9947:0509/114008.343527:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
```

Push to Docker Hub:

docker login

BACKEND:

docker push umeshchandra2001/k8s-assignment/python-backend:latest

docker tag umeshchandra2001/k8s-assignment/python-backend:latest
umeshchandra2001/k8s-assignment:latest

docker push umeshchandra2001/k8s-assignment:latest

FRONTEND:

docker build -t umeshchandra2001/k8s-assignment/flask-frontend:latest-frontend .

docker tag umeshchandra2001/k8s-assignment/flask-frontend:latest-frontend
umeshchandra2001/k8s-assignment:latest-frontend

docker push umeshchandra2001/k8s-assignment:latest-frontend

Apply Kubernetes manifests

kubectl apply -f mongo-deployment.yaml
kubectl apply -f mongo-express-deployment.yaml
kubectl apply -f backend-deployment.yaml
kubectl apply -f flask-deployment.yaml

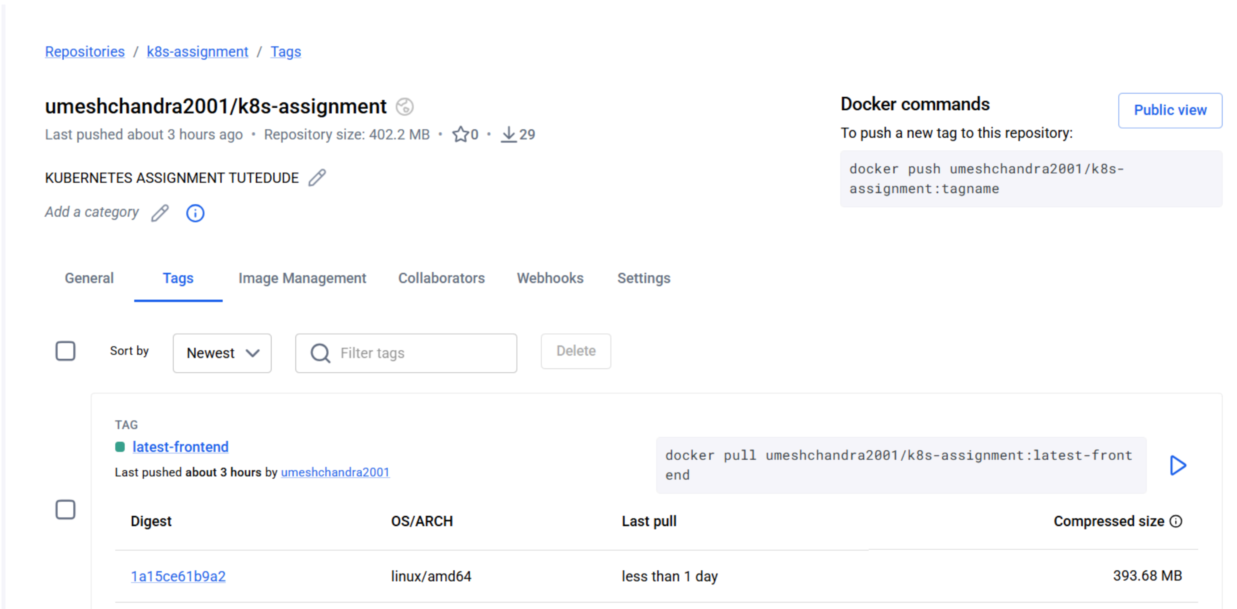
Get services

kubectl get svc

Checking in Dashboard:
Minikube dashboard

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$ kubectl apply -f flask-deployment.yaml
service/flask-service created
deployment.apps/flask created
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$ la -al
total 24
drwxrwxr-x 2 ubuntu ubuntu 4096 May  9 11:20 .
drwxrwxr-x 5 ubuntu ubuntu 4096 May  9 11:25 ..
-rw-rw-r-- 1 ubuntu ubuntu  508 May  9 14:50 backend-deployment.yaml
-rw-rw-r-- 1 ubuntu ubuntu  505 May  9 14:50 flask-deployment.yaml
-rw-rw-r-- 1 ubuntu ubuntu  572 May  9 14:52 mongo-deployment.yaml
-rw-rw-r-- 1 ubuntu ubuntu  756 May  9 14:52 mongo-express-deployment.yaml
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$ kubectl apply -f backend-deployment.yaml
service/backend-service created
deployment.apps/backend created
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$ kubectl apply -f mongo-deployment.yaml
service/mongo created
deployment.apps/mongo created
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$ kubectl apply -f mongo-express-deployment.yaml
service/mongo-express created
deployment.apps/mongo-express created
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$ kubectl get svc
NAME              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
backend-service   NodePort    10.105.209.1     <none>        4000:30003/TCP   2m50s
flask-service     NodePort    10.111.199.86    <none>        5000:30002/TCP   3m50s
kubernetes        ClusterIP   10.96.0.1        <none>        443/TCP          9m48s
mongo             ClusterIP   10.110.80.228    <none>        27017/TCP        61s
mongo-express     NodePort    10.108.201.72    <none>        8081:30001/TCP   47s
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/K8s-assignment/k8s$
```

```
File  Edit  View  Terminal  Tabs  Help
ubuntu@ip-172-31-16-42:~$ minikube start
😄  minikube v1.38.1 on Ubuntu 22.04
✨  Automatically selected the docker driver. Other choices: ssh, none
❗  Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
📌  Using Docker driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.35.1.
    ▪ Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-16-42:~$ minikube start
😄  minikube v1.38.1 on Ubuntu 22.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: default-storageclass, storage-provisioner

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.35.1.
    ▪ Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-16-42:~$ minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:44303/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
[9799:9799:0509/161507.205019:ERROR:base/memory/shared_memory_switch.cc:289] Failed global descriptor lookup: 7
Created TensorFlow Lite XNNPACK delegate for CPU.
[9851:7:0509/161507.807447:ERROR:gpu/ipc/client/command_buffer_proxy_impl.cc:287] ContextResult::kTransientFailure: Failed to send GpuControl.CreateCommandBuffer.

(google-chrome:9751): IBUS-WARNING **: 16:15:12.640: Unable to connect to ibus: Could not connect: Connection refused
[9751:9779:0509/161532.927382:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9751:9779:0509/161558.599254:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9751:9779:0509/161658.263547:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[9751:9779:0509/161828.033325:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
Fontconfig error: Cannot load default config file: File not found
```



PUSHING CODE GIT

```
git init
git add .
git commit -m "Initial commit - K8s assignment CRUD app"
git branch -M main
git remote add origin https://github.com/umeshchandra2001/K8s-assignment.git
git push -u origin main
```

CODE FILEAS HAVE PLACED IN GIT LINK FOR GIT REPOSITORY:

https://github.com/UmeshChandra2001/K8s-assignment

Submission guidelines: share github repo link and screenshots of your running deployments and commands.